# VF-C05 Taber Abrasion

| Test Method for Taber Abrasion | Test Method #: VF- C05<br>REF #: TM-R-039 |
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| Date Issued: Dec 2,2024<br>Revision #: 3 | Reason for revision:<br>The damage of the specimen must be checked in the center of the abraded area and not at the edges of it. |

**PURPOSE AND SCOPE:** To measure the abrasion resistance of the surface coating on leather and synthetic leathers using a rotary rubbing action under controlled pressure.

**REFERENCE: ASTM D 3884**

**EQUIPMENT AND MATERIAL**:

1. Taber abrasion machine, meeting the requirements of ASTM D3884 with rotary platform speed at 72 ± 2 revs/ min.
2. S-11,150 grade sandpaper B
3. S-36 cardboard backer A, 0.8 ± 0.2mm thick and 103 ± 3mm in diameter.
4. H-22 abrasive wheels
5. CS-17 abrasive wheels
6. H-18 abrasive wheels ***(TBL specification)***
7. Soft bristle brush.
8. Leather and synthetic leather.

**SAMPLE PREPARATION:**

1. Condition the sample for at least 24 hours at the standard laboratory environment @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Die cut test samples having a diameter of 103+/-3mm and should have a 7 ±1 mm hole in the center. ***Note: Hand cut specimen should not be used.***
3. The samples should be taken from different locations on the test material, and at least 10cm from any edge.
4. Adhere the backside of the test samples to a cardboard backer of the same diameter as the test sample.

**Note:** If a pressure-sensitive adhesive –backed cardboard is being used, adhere the test sample by simply pressing it onto the adhesive. If non-adhesive-backed cardboard is being used, adhere the test sample by applying a thin layer of adhesive of double-faced tape to the cardboard and then pressing the test sample onto the adhesive.

**EQUIPMENT PREPARATION:**

Equip the Taber machine with different abrasive wheels such as H-22, H-18 and CS-17. For **"Head Weights,** refer to individual weights to be used, and the number of cycle time per type of material as outlined on the **Material Specification Sheets**. The speed of rotary platform should be 72 ± 2 revs /min.

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**REPETITIONS**: Test two (2) samples per material.

**TEST PROCEDURE:**

***Test should be carried out at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.***

A. **Conditioning /Cleaning**

1. Place a new piece of 150 grade sandpaper on the turntable. Clamp the sandpaper in place with both the threaded center ring and the perimeter clamping ring.

2. Lower the abrasive wheels onto the sandpaper, then run the machine for 25 cycles. 3.

Raise the abrasive wheels from the turntable, remove the sandpaper and discard it.

4. Clean the abrasive wheels, using a soft bristle brush.

**Notes**:

- The abrasive wheels must be re-surfaced every 10-15 tests or when the surface becomes uneven or excessively clogged with debris.
- ***The abrasive wheels must be discarded when the diameter of the wheel is less than 44.5 mm.***
- ***The S-11 abrasive paper-refacing disc has a useful life 50 cycles, after which it must be discarded. Never reface wheels with a used refacing disc.***

**B. Testing**

1. Re-set the counter to zero.
2. Place the test sample on the turntable. Clamp the sample in place with both the threaded center ring and the perimeter clamping ring.
3. Lower the abrasive wheels onto the test sample and turn on the machine
4. During the first 25 cycles of the test, inspect the condition of the sample, then at 50 cycles, then at 250 cycles. If no severe failure has occurred at 250 cycles, check at 250 cycle intervals until failure or the maximum number of required cycles.
5. Failure is defined as the number of Taber cycles required to wear through the top finish, the damage of the specimen must be checked in the center of the abraded area and not at the edges of it.

**Note**: With full grain leather, this is also defined as the point when the surface finish is worn away such that the “peaks and valleys” of the grain cannot be distinguished.

**REPORT**: Number of cycles when failure observed.